

DEFINITION — HOW DATA IS DESCRIBED

API INTERFACES5



ODCS
BITOL @ LF



OpenAPI
LF



AsyncAPI
LF



GraphQL
LF



gRPC
CNCF

DATA PRODUCTS4



ODPS
BITOL @ LF



ODPSpec
LF



DPDS
ODM



DPROD
OMG

SCHEMA5



XML Schema
W3C



JSON Schema
IETF



SQL DDL
ISO/IEC



AVRO Schema
ASF



Protobuf
GOOGLE

SEMANTICS6



OSI
OSI INITIATIVE



RDF/OWL
W3C



DCAT
W3C



SKOS
W3C



SHACL
W3C



JSON-LD
W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS6



CSV
IETF



JSON
IETF



XML
W3C



PARQUET
ASF



AVRO
ASF



ORC
ASF

OPEN TABLE FORMATS3



Iceberg
ASF



Delta
LF



Hudi
ASF

STORAGE SYSTEMS2



S3
AWS



HDFS
ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY3



JDBC
JCP



ODBC
ISO/IEC



ADBC
ASF

INTERCONNECTION2



HTTP
IETF



FTP / SFTP
IETF

MESSAGING4



Kafka
ASF



AMQP
OASIS



MQTT
OASIS



CloudEvents
CNCF

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2



Apache Arrow
ASF



DataFrame API
DATA-APIs

PROCESSING4



dbt
DBT LABS



Spark
ASF



Pandas
NUMFOCUS



Ibis
IBIS

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS5



Iceberg Catalog
ASF



Unity Catalog
LF



Hive Metastore
ASF



DuckLake
DUCKDB LABS



Schema Registry
CONFLUENT

LINEAGE1



OpenLineage
LF

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY2



SQL
ISO/IEC



SPARQL
W3C

DATA QUALITY3



dbt
DBT LABS



Great Expectations
GX LABS



SodaCL
SODA

OBSERVABILITY2



Open Telemetry
CNCF



OORS
BITOL @ LF

POLICIES1



OPA
CNCF

AI INTERFACES2



MCP
LF



A2A
LF

☐ Open standard ☒ Highlighted by Entropy Data — the ones to bet on today ☐ Emerging / draft ☐ Single-vendor — muted